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First Paleomagnetic Result From the Early Permian Volcanic Rocks in Northeastern Mongolia: Evolutional Implication for the Paleo-Asian Ocean and the Mongol-Okhotsk Ocean

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Key Points:
- An Early Permian paleomagnetic pole is reported from northeastern Mongolia
- The Mongolia Block should have welded with the North China Craton before the Early Permian
- The Mongolia-North China Block was separated from the Siberia Craton by the Mongol-Okhotsk Ocean with ~30° latitudinal difference in Early Permian

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Abstract The Mongolia Block (MOB), which is now sandwiched by the Siberia Craton (SIB) and the North China Craton (NCC), plays an essential role for understanding the late stage evolution of the Paleo-Asian Ocean and the early stage evolution of the Mongol-Okhotsk Ocean. Here, a paleomagnetic study is performed for the first time on the Early Permian volcanic strata in the Bayandun region of northeastern Mongolia and the data are used to uncover the late Paleozoic paleoposition of the MOB and better understand the evolution of both oceans. Zircon U-Pb dating results reveal an emplacement age of 283 ± 3 Ma for the studied volcanic strata. Rock magnetic analyses identify that titanium-poor magnetite is the main magnetic carrier. Characteristic remanent magnetization isolated from seven sites shows consistent reverse polarity, corresponding to the Permo-Carboniferous (Kiaman) Reverse Superchron. Site-mean directions pass fold tests, and an Early Permian paleomagnetic pole is calculated for the MOB at $\lambda/\varphi = -14.9°\text{N}/76.8°\text{E}$ ($A_{95} = 5.7°$) with $N = 7$ sites. Comparison with published Permian paleomagnetic poles from surrounding blocks indicates that (1) the MOB should have welded with the NCC before the Early Permian or was at least very close to it. (2) The welded MOB-NCC was separated from the SIB by the Mongol-Okhotsk Ocean with ~30° latitudinal difference during the Early Permian. (3) Significant vertical-axis strike-slip related rotations occurred within and along the margins of the unified MOB-NCC due to the far-field stress effect produced by posterior orogenic events.

1. Introduction

Central and north Asia was constructed by amalgamation of three main blocks, the Siberia Craton (SIB), the Mongolia Block (MOB; also referred to as the Amuria Block), and the North China Craton (NCC), from north to south (Figure 1; Enkin et al., 1992). Welding of these three blocks has been achieved by closure of two major oceans: the Mongol-Okhotsk Ocean between the SIB and the MOB (Van der Voo et al., 2015; Wu et al., 2017) and the Paleo-Asian Ocean between the NCC and the MOB (Xu et al., 2013; Zhao et al., 2013). The MOB, geographically occupying eastern and southern Mongolia, Amur region of Russian Far East, and northeastern China, is sandwiched by the SIB to the north and the NCC to the south, with the Mongol-Okhotsk suture zone and the Solonker suture zone as boundaries (Xiao et al., 2003; Zonenshain et al., 1990). Therefore, the paleoposition of the MOB in different periods is essential to understand the evolution of both oceans and timing for formation of two suture zones. The evolution of both oceans has become one of the most controversial tectonic problems of northern Asia in the last decades. For the closure timing of the Paleo-Asian Ocean, the Late Devonian and Late Permian ages have been proposed based on sedimentological, paleontological, geochemical, paleomagnetic, and structural studies (Xiao et al., 2003; Xu et al., 2013; Zhao et al., 2013; Zhang et al., 2018). A scissor-style gradual closure model for the Mongol-Okhotsk Ocean from Triassic in the western segment to latest Jurassic-earliest Cretaceous in the eastern segment has been interpreted on the basis of geological and paleomagnetic studies (Zhao et al., 1990; Zorin, 1999; Kravchinsky, Sorokin, & Courtillot, 2002, Kravchinsky, Cogné, et al.,

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2002; Metelkin et al., 2010). However, west-east simultaneous closure of the Mongol-Okhotsk Ocean during the latest Jurassic-earliest Cretaceous has also been proposed based on a compilation of paleomagnetic data (Van der Voo et al., 2015).

In the last few decades, several paleomagnetic studies have been carried out in Mongolia. However, due to the insufficient paleomagnetic data and contradicting conclusions deduced from these data, the Late Paleozoic paleoposition of the MOB and its relationship to the SIB and the NCC are still highly debatable. Pruner (1987, 1992) performed the first paleomagnetic study on the Carboniferous and Permian strata in northern and central Mongolia, and the data showed a low (~20°N) paleolatitude for the MOB, which is consistent with that of the NCC. Therefore, the author suggested that central and northern Mongolia had probably been a part of the NCC during Permian, and far away (~40° distance) from the SIB (Pruner, 1987, 1992). Kovalenko and Chernov (2008) reported an Early Permian pole from the Daribi Ridge in western Mongolia. However, this pole is calculated from number of samples rather than sites and therefore no fold test can be applied. Moreover, the data show very high paleolatitudes (45–60°N), which enabled the authors to suggest that western Mongolia was located at the southern margin of the SIB and separated from the NCC by an ocean with width of ~2,000 km (Kovalenko & Chernov, 2008). As the samples were collected from the southern part of the Tuva-Mongolia microblock that accreted to the SIB during the Neoproterozoic-Early Paleozoic, this pole may not be able to represent the MOB. Edel et al. (2014) reported paleomagnetic data from Carboniferous volcanic rocks and granodiorites in southwestern Mongolia. However, as no fold test was applied, the components isolated are either remagnetized or have an unclear age of remanence acquisition.

Therefore, up to now, reliable Late Paleozoic paleomagnetic poles are still rare for the MOB. Paleomagnetic results from well-dated strata are needed to constrain the paleogeographic evolution of the MOB in the Late Paleozoic and to further understand the evolution of both the Paleo-Asian Ocean and the Mongol-Okhotsk Ocean. In order to achieve these goals, we carried out paleomagnetic studies on the Early Permian volcanic strata in northeastern Mongolia.

2. Geological Background and Sampling

In northeastern Mongolia, the core MOB is constituted by the Precambrian Ereendavaa Terrane, which extends eastward into northeastern China and Russian Far East (Figure 1; Badarch et al., 2002). Radiometric age data are rare for this terrane in the Mongolian part. However, in the Russian part, zircon U-Pb dating of a biotite granite sample gave an age of 740 ± 20 Ma (Bibikova et al., 1979). In addition, granitoids and gabbros with ages of 851 ± 6 Ma to 737 ± 5 Ma have been identified along the Argun River in northeastern China (Tang et al., 2013). The Precambrian basement is overlain by Neoproterozoic metasedimentary and metavolcanic rocks and intruded by granitoids with Cambrian to Devonian ages (Badarch et al., 2002). Ordovician-Silurian sandstone and limestone have been widely identified along its southern margin. All the pre-Permian rocks are intruded by Permian-Triassic granitoids (Badarch et al., 2002). In its northern part, a Permian volcanic sequence and a Triassic-Jurassic sedimentary sequence were identified (Donskaya et al., 2013). Triassic-Jurassic strata along its northern margin experienced severe ductile deformation due to the Mongol-Okhotsk orogeny.

The Early Permian Gazar Formation is well exposed in the Bayandun region to the south of the Mongol-Okhotsk suture (Figures 1 and 2). This volcanic-dominated sequence is mainly composed of rhyolite, dacite, andesite, and tuff, with subordinate siltstone and sandstone interlayers. Due to posterior tectonic events, this formation shows highly tilted or even vertical beddings with tight folds (Table 1). The age of this volcanic sequence has been assigned to the Early Permian during previous geological mapping. However, there are no fossil or radiometric data; thus, the age of this formation is not well constrained. Therefore, we collected one rhyolite sample for zircon U-Pb dating to determine the age of this volcanic sequence, which will also contribute to our paleomagnetic analysis and for comparison with coeval paleomagnetic poles from nearby blocks.

Overall, 69 cores of nine sites were drilled from the Early Permian Gazar Formation in northwestern Bayandun (Figure 2 and Table 1). Sites 08-13 were sampled from section A, and sites 14–16 were collected from section B (1.5 km to the northeast of section A). Sections A and B represent two limbs of a fold

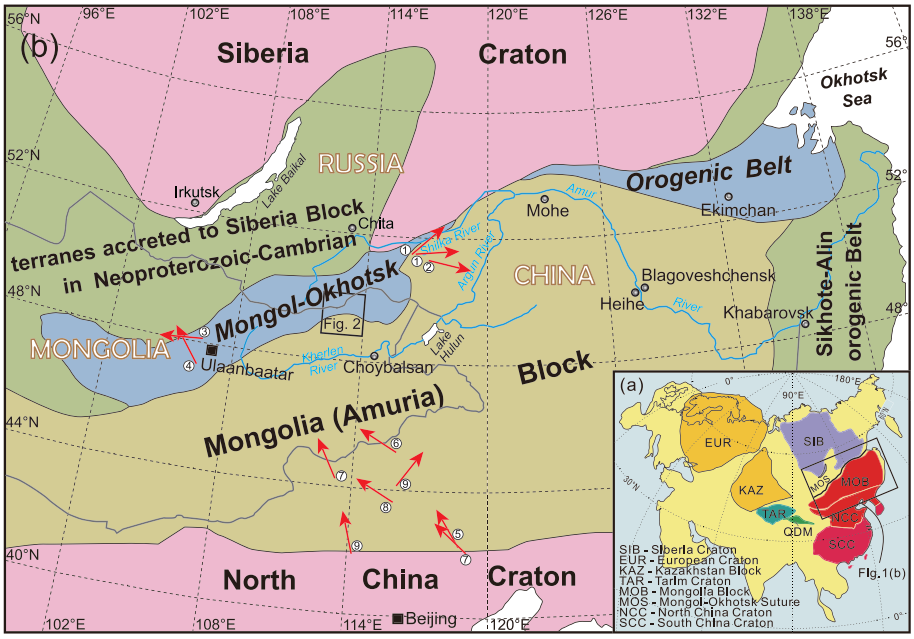


Figure 1. (a) Schematic map of Asia with major Precambrian blocks. (b) Simplified tectonic map showing the Mongolia Block between the Siberia Craton and the North China Craton, with the highlighted Mongol-Okhotsk suture zone extending from central Mongolia to the present-day Okhotsk Sea (modified from Zhao, Xu, & Jahn, 2017). Arrows represent Permian paleomagnetic directions obtained in this region (data in Russia are from ①Xu et al., 1997 and ②Kravchinsky, Cogné, et al., 2002; data in Mongolia are from ③Pruner, 1987 and ④Pruner, 1992; data in northern China are from ⑤Zhao et al., 1990; ⑥Chen et al., 1997; ⑦Li et al., 2012, ⑧Zhao et al., 2013 and ⑨Zhang et al., 2018).

(Table 1), which can be used for a fold test of our paleomagnetic data. Six to nine cores were obtained from each site using a portable gasoline drilling machine. Each core was orientated in the field by both magnetic and sun compasses.

3. Analytical Procedures

3.1. Zircon U-Pb Dating

Zircon grains were initially selected by heavy liquid separation and then purified with a magnetic separation technique. The grains left were then handpicked under a binocular microscope. Picked zircons were embedded in an epoxy resin and half-sectioned. All grains mounted were photographed under an optical microscope with both reflected and transmitted light to reveal surface flaws and internal inclusions. Finally, they were subjected to cathodoluminescence (CL) to reveal internal textures enabling us to choose target spots for further U-Pb dating. The CL imaging was carried out with a Quanta 200 FEG Scanning Electron Microscope in the SEEL lab of Peking University. Zircon U-Pb analysis was performed at the Key Laboratory of Orogeny and Crust Evolution (Peking University) using a LA-ICP-MS system with a 193-nm ArFExcimer laser and an Agilient 7500c ICP-MS. NIST 610 and Si were used as external and internal standards to calibrate zircon analyses. Zircon standards Plesovice (337 Ma) and 91500 were used to correct U-Pb isotope fractionation effects and to monitor the age measurement deviation, respectively. Calculation of element concentrations and isotopic ratios was done with the GLITTER software (ver. 4.4.2) designed by Macquarie University. For common lead correction, we follow Andersen's (2002) method. Concordia ages were calculated by Isoplot/Excel (3.0), with the weighted mean ages at 95% confidence level (Ludwig, 2003).

3.2. Paleomagnetism

Cores were cut into 2.2-cm-long standard specimens in the laboratory. Before magnetic remanence measurements, several investigations were performed to determine magnetic carriers: high-temperature thermomagnetic runs of magnetic susceptibility (κ-T) using a CS3 furnace-coupled KLY-3S kappabridge, acquisition

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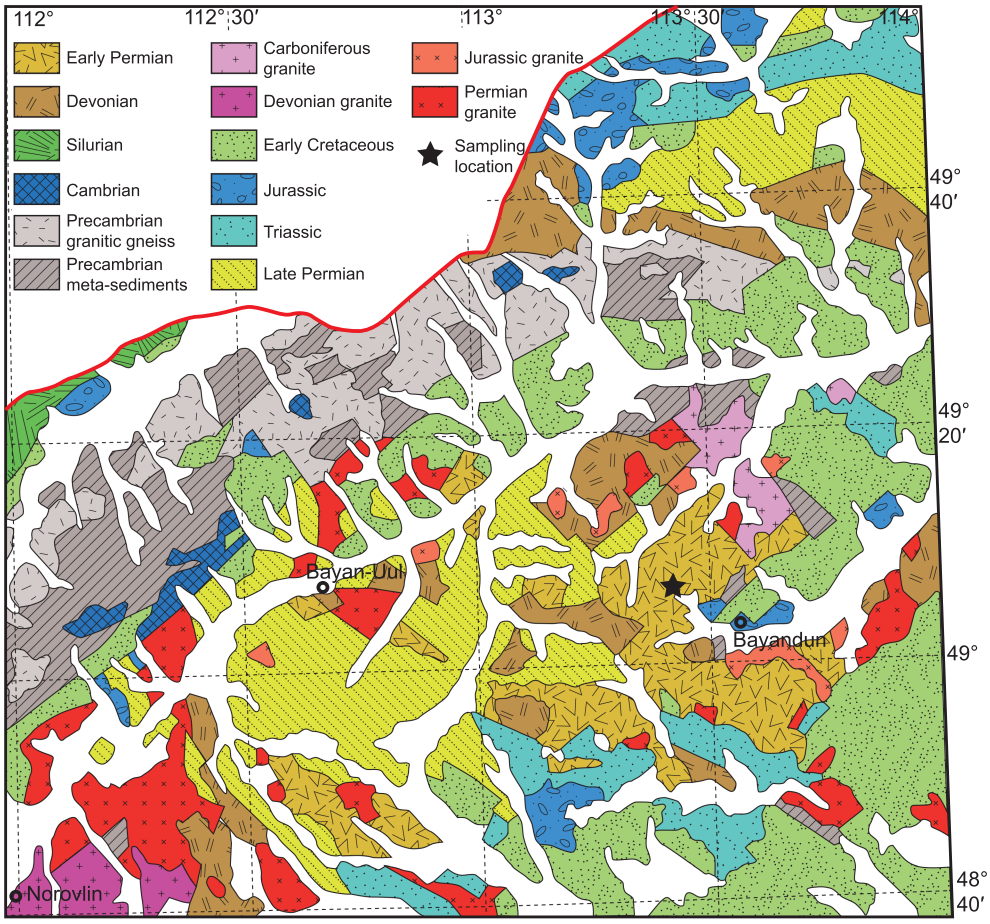


Figure 2. Geological map of the Bayandun region showing the paleomagnetic sampling location (from Mineral Resources Authority of Mongolia & Mongolian Academy of Science, 1998).

curve of isothermal remanent magnetization (IRM) by imparting IRMs with a MMPM9 pulse magnetizer and measuring IRM intensities with a Minispin spinner magnetometer, and magnetic hysteresis curve measurements using a Princeton Measurements MicroMag 2900 Alternating Gradient Magnetometer (AGM). All these rock magnetic analyses were done in the magnetic laboratory of University Tübingen, Germany.

For each site, six to eight specimens were chosen for demagnetization. Magnetic remanence was progressively removed in 13 to 16 steps using both thermal and alternating magnetic field (AF) techniques. The temperature intervals are from 20 to 100 °C, and the alternating magnetic field intervals are from 2 to 20 mT. For thermal treatment, a model TD-48SC thermal demagnetizer (ASC Scientific Company) was used. Magnetic remanences were measured with a 2G Enterprises u-channel DC-SQUID (Model 755–1.65 UC) magnetometer. A degausser is implemented in the magnetometer system for alternating field demagnetization.

Magnetic directions were calculated using principal component analysis method (Kirschvink, 1980). Fisher spherical analysis was used to compute mean magnetic directions. Individual direction isolation, mean direction computation, and pole calculation and comparison were achieved by the softwares Paleomag (Cogné, 2003), PMGSC (designed by R. Enkin), and Remasoft (designed by AGICO).

4. Analytical Results

4.1. Zircon U-Pb Age

Zircons of the rhyolitic sample (M15-82) show well-developed euhedral prismatic shape and oscillatory growth zoning (Figure 3). Uranium concentrations range from 270 to 1,656 ppm, and Th/U ratios range from 0.44 to 1.00 (supporting information Table S1). These characteristics indicate a magmatic origin of dated

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Table 1
Paleomagnetic Sampling Sites and Measurement Results From Northeastern Mongolia

Site	Coordinate	Rock type	Age	Azimuth of bedding dip direction/dip	n′/n	N/R	Dg	Ig	Ds	Is	k	α_{95}	Comment
Early Permian volcanic rocks													
08	49°06′50.0″N,113°24′52.1″E	rhyolite	P_1gz	245/89	8/8	7/1	8.6	−53.8	33.8	18.2	2.2	50.3	high scatter
09	49°06′50.3″N,113°24′51.7″E	dacite	P_1gz	245/89	6/6	0/6	266.3	−46.9	46.0	−40.4	31.9	12.1	
10	49°06′50.2″N,113°24′51.6″E	decite	P_1gz	245/89	6/8	0/6	277.2	−49.8	40.4	−34.0	20.3	15.2	
11	49°06′50.2″N,113°24′51.5″E	dacite	P_1gz	245/89	6/6	0/6	278.0	−52.0	41.7	−32.1	62.9	8.5	
12	49°06′50.4″N,113°24′52.8″E	dacite	P_1gz	245/89	8/9	0/8	276.7	−47.4	38.9	−36.1	52.3	7.7	
13	49°06′50.3″N,113°24′51.5″E	dacite	P_1gz	245/89	7/7	0/7	276.7	−44.4	36.3	−38.4	128	5.4	
14	49°07′29.4″N,113°25′36.6″E	dacite	P_1gz	38/80	4/6	0/4	25.6	44.9	27.4	−34.0	26.4	18.2	
15	49°07′28.5″N,113°25′36.4″E	dacite	P_1gz	38/80	6/8	2/4	333.5	52.7	8.6	−4.8	1.4	100.8	high scatter
16	49°07′27.0″N,113°25′34.9″E	dacite	P_1gz	38/80	3/8	0/3	26.6	60.8	32.2	−18.6	138	10.5	
mean (7 site)							**296.0**	**−34.5**			**1.9**	**63.6**	
									37.4	**−33.5**	**85.8**	**6.6**	

Note. Abbreviations. n′/n: numbers of accepted/measured samples; N/R: normal/reverse polarity; Dg, Ig, Ds, Is: declination (D) and inclination (I) in geographic (g) and stratigraphic (s) coordinates; k: precision parameter; α_{95}: 95% confidence limit.

zircons. Twenty zircons were analyzed, and after rejection of discordant and ungrouped data 12 age points yield $^{206}Pb/^{238}U$ age at 283 ± 3 Ma (MSWD = 1.7), which we interpret as the extrusion age for this volcanic sequence (Figure 3).

4.2. Paleomagnetism

4.2.1. Rock Magnetism

Thermomagnetic (κ-T) curves show gradual decreasing values of susceptibility during heating with a Curie temperature near 580 °C indicating magnetite (Figures 4a and 4b). The decay above ~600 °C in some curves (e.g., Figures 4a and 4b) indicates the existence of a substantial amount of hematite. Enhancement of susceptibility values at ~580 °C in the cooling curves suggests formation of new minerals (most likely magnetite) during the heating process above 580 °C (Figures 4a and 4b). IRM acquisition curves reveal a rapid increase below 200 mT and complete saturation at ~300 mT (Figure 4c), indicating the dominance of soft magnetic

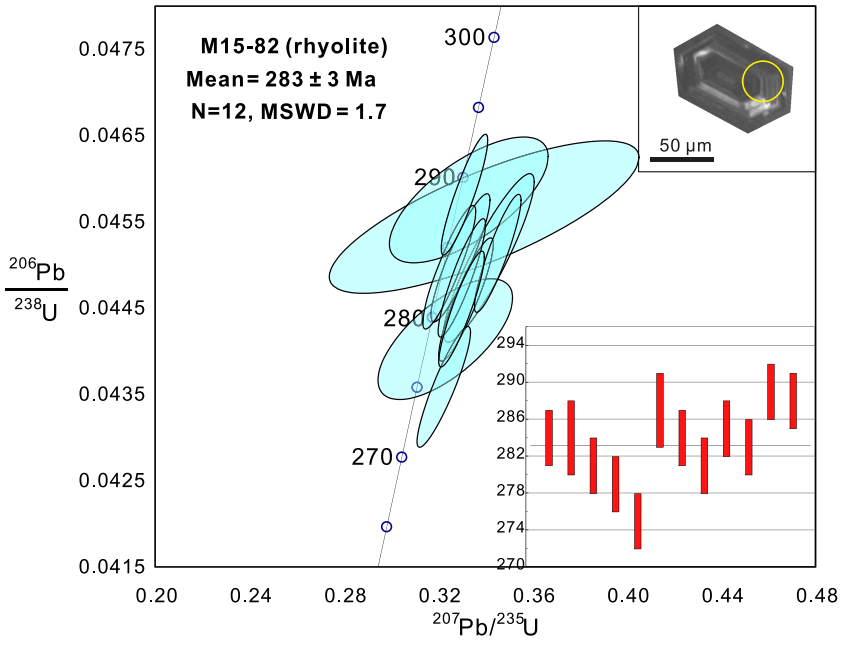


Figure 3. U-Pb concordia diagram and weighted mean age plot of rhyolite samples. The inset picture shows CL image of representative zircon grain.

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minerals in these samples. In some samples a small increase at higher fields is observed (Figure 4d). Hysteresis loops generally show narrow shapes and tend to close before 200 mT with low-coercivity values around 2.5–3 mT (Figures 4e and 4f), suggesting that soft magnetic minerals (e.g., magnetite or Ti-magnetite) are predominant in our samples. Meanwhile, thermal demagnetization shows unblocking temperature at ~510 °C (Figure 5), indicative of Ti-magnetite. Therefore, magnetite and Ti-magnetite can be considered as the main magnetic carriers in the studied volcanic samples.

4.2.2. Magnetic Directional Analysis and Paleomagnetic Pole

The specimens were demagnetized by both thermal and AF treatment. In order to check the consistency of results, several cores were cut into two specimens for thermal and AF demagnetization. Results from both methods reveal consistent behavior and similar magnetic directions (Figures 5d, 5e, 5i, and 5j). Most specimens display linear magnetization direction with maximum angular deviation less than 5° for most samples. One or two components were isolated from the orthogonal vector plots (Figure 5). Low-temperature (0–200 °C)/soft (0–20 mT) components isolated from 20 specimens show dispersed remanence directions in both geographic and stratigraphic coordinates (Figure 6). The meaning of this low-temperature/soft component is not clear. After removing the low-temperature/soft component, most specimens show a characteristic component with rectilinear decay to the origin, which is successfully isolated by both thermal and AF treatments (Figure 5). This component can be considered as the characteristic remanent magnetization (ChRM), which is tightly clustered for most sites, except for sites 08 and 15, which are rejected due to high scatter ($k = 2.2$ and 1.4, respectively; Table 1). Before tilt correction, sites 09–13 from section A and sites 14–16 from section B show clear different directions (Table 1 and Figure 6). After tilt correction, all samples display similar directions of reverse polarity (Figures 5 and 6), which is consistent with the Permo-Carboniferous (Kiaman) Reverse Superchron (Opdyke & Channell, 1996).

Seven site-mean directions were included to calculate the formation mean direction of $D_g/I_g = 296.0°/-34.5°$ ($k_g = 1.9$, $\alpha_{95g} = 63.6°$) in geographic coordinate and $D_s/I_s = 37.4°/-33.5°$ ($k_s = 85.8$, $\alpha_{95s} = 6.6°$) after tilt correction (Figure 6). Site-mean directions are significantly clustered after tilt correction with maximum k value at around 105% untilting (Figure 6). They pass McElhinny's (1964) fold test at 99% confidence level, and direction-correction (DC) fold test of Enkin (2003) also gives a positive feedback with DC slope at 97.5% ± 7.0% untilting (Figure 6). The stepwise correction modeling (Watson & Enkin, 1993) shows the optimal degree of unfolding at 105.2% ± 7.3%, indicating a prefolding acquisition of the magnetic remanence. The age of this folding is most likely middle Jurassic to earliest Cretaceous, when the MOB collided with the SIB. The unified reverse polarity for all samples indicates that the magnetic remanence was acquired in the period of the PCRS (Opdyke & Channell, 1996). Meanwhile, this formation-mean direction is different from Triassic and Jurassic paleomagnetic directions reported from the MOB (Pruner, 1992; Zhao et al., 1990). Therefore, we can conclude that the ChRM of the studied volcanic rocks is a primary magnetization acquired during the Early Permian. A corresponding Virtual Geomagnetic Pole (VGP) is calculated at $\lambda/\varphi = -14.9°N/76.8°E$ ($A_{95} = 5.7°$) with $N = 7$ sites, which represents the Early Permian paleomagnetic pole for the MOB (Table 1).

5. Discussion

5.1. Comparison of Permian Paleomagnetic Poles

Paleosecular variation (PSV) is a fundamental problem for paleomagnetic data from volcanic rocks, as to whether the PSV has been averaged out or not, and will influence the reliability of the data set. Therefore, before comparison of paleomagnetic poles, we first examined the PSV problem of our data set. In order to average out secular variation, we sampled from two sections. For the data set, the Fisher precision parameter k values of site-mean directions are mostly in the range of 20–63 (Table 1), indicating relative scatter of paleomagnetic directions for individual specimen within each site. Moreover, the A_{95} value (5.7°) of the VGP is lower than the A_{95min} value (7.8°) calculated after Deenen et al. (2011), indicating that PSV may have already been averaged out within each site. However, the angular dispersion S (= 8.7) of the VGPs of the seven sites is slightly lower than that of expected PSV from the TK03.GAD model at the same latitude (Tauxe & Kent, 2004), indicating that the PSV is not fully averaged out. Even though our data might not be adequate to fully average out secular variation, they support previous paleolatitude estimate of the MOB without the problem of secular variation. Therefore, we can still use this pole for comparison and tectonic discussion.

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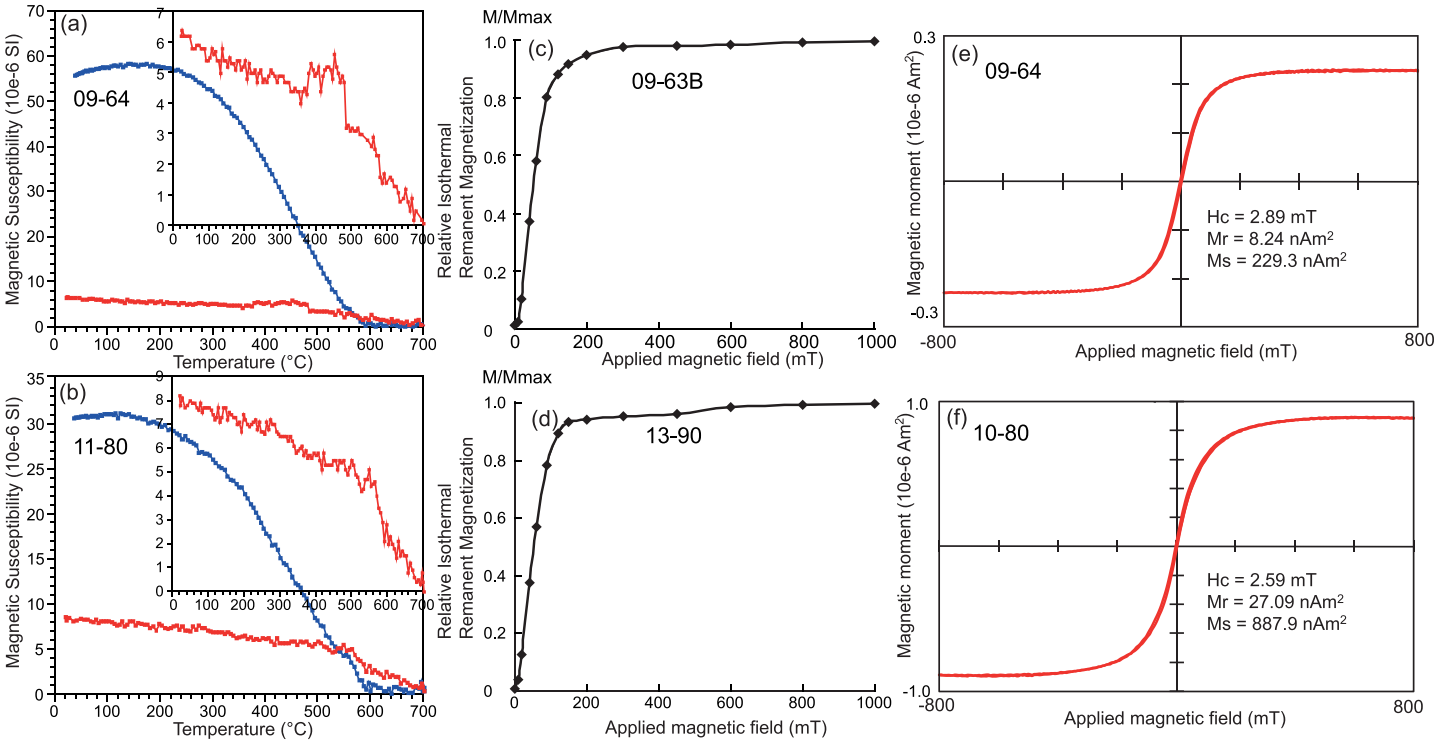


Figure 4. Magnetic mineral analyses. Thermomagnetic curves (a, b; red/blue represent heating/cooling), isothermal magnetization acquisition curves (c, d), and magnetic hysteresis loops (e, f) of selected samples.

Late Paleozoic, especially Permian, strata are widely distributed in northeastern Mongolia, which provide a good opportunity for paleomagnetic study. Pruner (1987, 1992) reported an Early Permian and a Late Permian paleomagnetic pole from northern and central Mongolia, but no fold test could be applied for either of these (Table 2 and Figure 7). Although both normal and reverse polarities have been found in the Early Permian strata, specimens with normal polarity account for 74% of the total specimens (Pruner, 1992), which is inconsistent with the PCRS. Therefore, this Early Permian pole might be questionable. Pruner's data (1987, 1992) show a low paleolatitude (11–21°N) for Mongolia during the Permian. Our data provide the first Early Permian paleomagnetic pole from NE Mongolia, which also indicates a low paleolatitude (~18.3°N) for Mongolia. Three Permian poles were reported from the Chita area of Russia (Figure 7; Xu et al., 1997; Kravchinsky, Cogné, et al., 2002), which is considered as the northern margin of the MOB. These three poles also correspond to a low (~ 20°N) paleolatitude similar to the data obtained from Mongolia (Figure 8).

For the southern part of the MOB, five paleomagnetic poles have been reported from Inner Mongolia in northern China (Table 2; X. Zhao et al., 1990; Chen et al., 1997; Li et al., 2012; P. Zhao et al., 2013; Zhang et al., 2018). Poles obtained by Chen et al. (1997) and Li et al. (2012) are both from the Baoligaomiao Formation in northern Inner Mongolia. The similar bedding attitude makes fold tests impossible for either of them. Although both normal and reverse polarities have been isolated by Chen et al. (1997), the normal and reverse magnetizations are not exactly antipodal, with about 20° deviation in declination and samples with normal polarity show a larger scatter (Chen et al., 1997). The authors interpreted that the discrepancy is probably caused by late Mesozoic overprint that has not been completely cleaned (Chen et al., 1997). Therefore, this pole might be questionable and we will not use it for further discussion. Samples from Li et al. (2012) show unified reverse polarities, which are consistent with the PCRS. The pole calculated by Li et al. (2012) is close to the Permian pole of Inner Mongolia (Zhao et al., 1990). Therefore, we may tentatively consider that the C_2-P_1 pole obtained from the Baoligaomiao Formation by Li et al. (2012) is primary, which gives a low paleolatitude (~18°N) for the southern MOB. Zhao et al. (1990) reported a middle Permian pole from eastern Inner Mongolia. It passes the fold test and is close to Permian poles from the NCC (Figure 7). Zhao et al. (2013)

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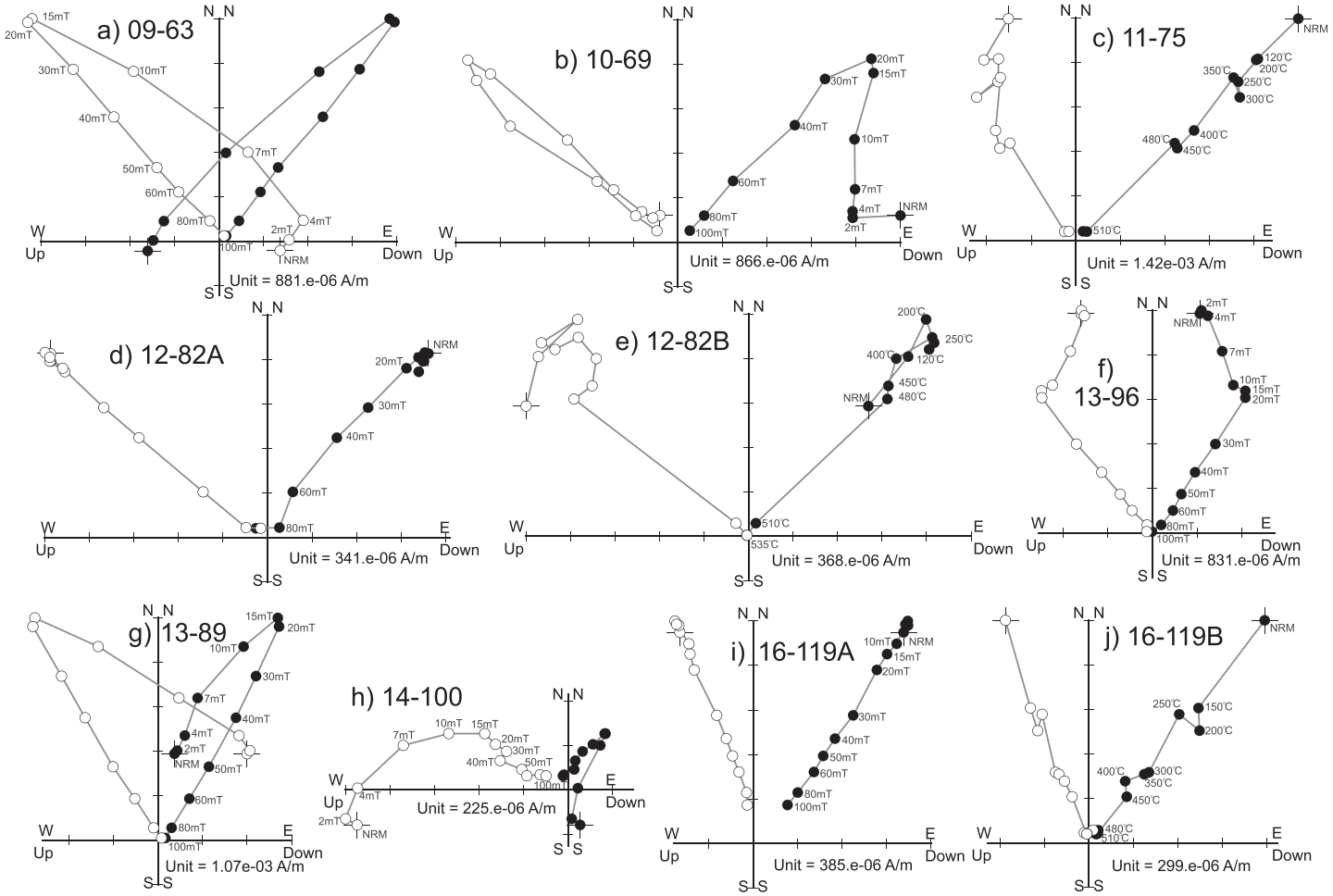


Figure 5. Typical orthogonal vector plots (Zijderveld, 1967). Directions are plotted in tilt-corrected coordinates. Black/white circles represent vector endpoints projected onto the horizontal/vertical plane. Numbers on the plots show temperature/applied alternating magnetic field steps. NRM: natural remanent magnetization.

obtained a new Permian paleomagnetic pole from central Inner Mongolia, which is overlapping with poles from the NCC (Figure 7), with indistinguishable angular (2.6° ± 10.9°) and paleolatitudinal (5.8° ± 8.5°) differences, indicating no relative movement between the NCC and the southern MOB during Permian (Zhao et al., 2015). The overlapping of paleomagnetic poles (Figure 7) and consistent low paleolatitudes of the southern MOB and the NCC (Figure 8) indicates that the southern MOB should have welded to the NCC during Permian or was at least close to it (Li et al., 2012; Pruner, 1987, 1992; Zhao et al., 2013). Recently, Zhang et al. (2018) performed paleomagnetic studies on the Early Permian volcanic strata in Inner Mongolia, which gave an equatorial paleolatitude (2.9°S–4.9°N). This unusual paleolatitude is different from previous reported paleomagnetic data in Inner Mongolia (Figure 8). Meanwhile, directions of the characteristic remanence obtained in Zhang et al.'s (2018) study are nearly perpendicular to previous ones (Figure 1b), which lead us to suspect that local deformation caused declination changes.

Although all Permian paleomagnetic data show a low paleolatitude for the MOB, these Permian paleomagnetic poles are dispersed in the equal-area projection (Figure 7). A small circle was calculated centered on the sampling location (49.1°N, 113.4°E), and this small circle intercepts all 12 Permian poles from the MOB including the Early Permian pole obtained in this study (Figure 7). This distribution indicates that regional strike-slip movement and vertical axis rotation should have occurred within or along the margins of MOB (e.g., Metelkin et al., 2010; Zhao et al., 2013, 2015).

The Permian poles of the NCC are consistent with each other, except for the pole of Wu et al. (1993), which is ungrouped due to local strike slip. Meanwhile, poles from the NCC are intercepted by the small circle

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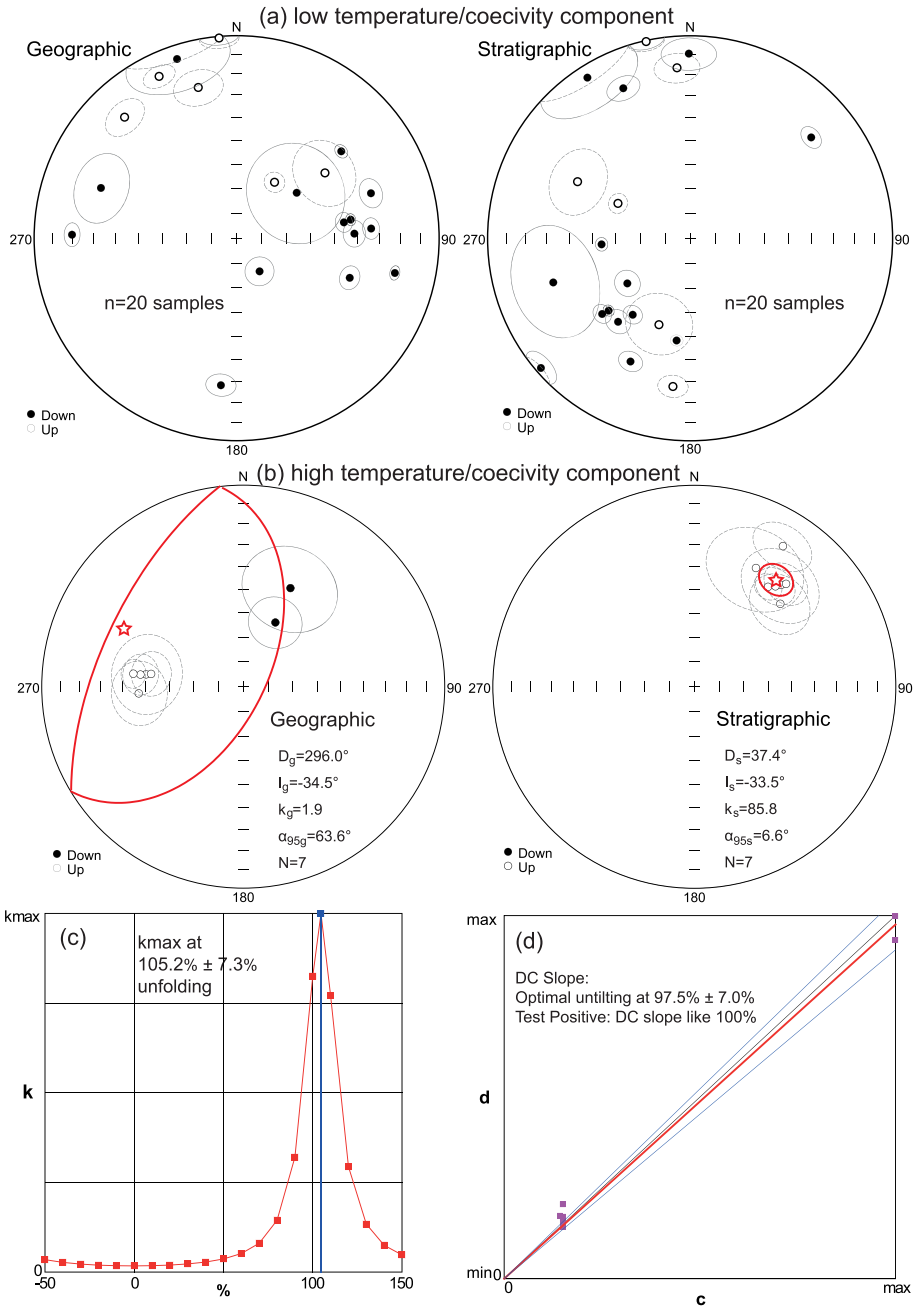


Figure 6. Equal-area projection of (a) low-temperature/coercivity component from 20 specimens and (b) site-mean directions of high temperature/coercivity component of seven sites in geographic and stratigraphic coordinates for the Early Permian Gazar Formation. Stars show overall site-mean direction, solid circles represent normal polarity, while open circles represent reverse polarity. (c) Progressive unfolding of site-mean directions shows a maximum directional clustering at ~105% bedding untilting. (d) Enkin (2003) DC fold test gives a DC slope at ~98% untilting, indicating a positive test.

calculated for the MOB. This observation indicates a similar paleolatitude of these two blocks during Permian and significant post-Permian strike-slip movements of the MOB relative to the NCC (Zhao et al., 2013, 2015).

The Permian poles of the SIB are rather consistent (Figure 7), and they are far away from those obtained from the NCC and the MOB, indicating a significant latitudinal distance between the SIB and the NCC-MOB during Permian. The paleolatitudes calculated for the SIB are around 50°–80°N during Permian, which is much

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Table 2
Compilation of Permian Paleomagnetic Poles From the North China Block, Amuria Block, and Siberia Block

Age	Slat (°N)	Slong (°E)	N	Plat (°N)	Plong (°E)	A_{95}(°) (dp/dm)	Test	References
North China Block (NCB)								
P_2	39.6	98.0	33s	42.4	350.9	3.9	F	Meng et al. (1990)
P_2	35.5	110.3	14S	49.5	358.7	2.5/4.6	F+R	Ma and Yang (1993)
P_2			11L	50.3	355.2	5.7	F	Yang et al. (1998)
P_2	37.8	112.3	12S	44.0	358.0	6.9	F	McElhinny et al. (1981)
P_2	37.5	114.4	4S	47.1	356.9	7.7	F	Zhao and Coe (1989)
$P_{1\text{-}2}$	42.0	113.9	24S	66.2	328.1	4.1	F	Zhang et al. (2018)
P_1	35.5	110.3	4S	47	359.2	3.1/5.7	R	Ma and Yang (1993)
P_1	37.6	101.3	5S	23.9	21.2	12.8/21.4	F	Wu et al. (1993)
latest C_2	42.5	119.3	4S	53.1	355	4.8	R	Li et al. (2012)
Mongolia Block (MOB)								
Inner Mongolia (China)								
P_2	42.7	118.9	11S	52.5	339.5	6.3	F	Zhao et al. (1990)
P_1-P_3	43.7	118.4	6S	48.7	3.7	5.2/9.1	F	Zhao et al. (2013)
P_1	44.3	116.1	14S	41.6	246.1	7.5	F	Zhang et al. (2018)
C_2-P_1	45.5	117.2	5S(25s)	53.4	14.8	9.7	-	Chen et al. (1997)
C_2-P_1	44.3	113.3	7S	58.5	333.4	5.2	-	Li et al. (2012)
Mongolia								
P_3	48.1	105.9	27s	11.3	35.2	5.4/9.1	-	Pruner (1987)
P_1	47.8	107.1	4S	44.8	335.1	11.6	R	Pruner (1992)
P_1	49.1	113.4	7S	−14.9	76.8	5.7	F	This study
Amur river region (Russia)								
P_2	51.5	115.4	14S	20.5	200.6	14.5	F	Xu et al. (1997)
P_2	50.7	116.9	14S	8.3	183.9	9.5/16.2	F	Kravchinsky, Cogné, et al. (2002)
P_1	51.5	115.4	5S	33.8	207.8	26.3	F	Xu et al. (1997)
Siberia Block (SIB)								
P_3(250)	64.6	114.7	9L	50.8	149.6	9.4	-	Kravchinsky, Konstantinov, et al. (2002)
P_3(250)			7L	56.2	151.7	3.8	-	Veselovsky et al. (2003)
P_3(251)	67.5	104.0	8S	55.1	147.0	5.0	-	Pavlov et al. (2007)
P_3(251)	67.0	89.0	8S	57.2	151.1	4.0	-	Pavlov et al. (2007)
P_3	50.8	107.2	15S	63.1	151.0	13.5/15.0	F	Kravchinsky, Cogné, et al. (2002)
P_1(275)	51.7	104.1	5S	50.5	121.4	16.9	BC	Pisarevsky et al. (2006)

Note. Abbreviations: N, number of locations (L), sites (S), or samples (s) used for calculation; Slat (Plat), latitude of sampling site (pole); Slong (Plong), longitude of sampling site (pole); A_{95}, 95% confidence limit; F, fold test; R, reversal test; BC, baked contact test.

higher than those of the NCC and the MOB (Figure 8). This difference leads to the proposal that the MOB and the NCC were far away (~40–60° latitude distance) from the SIB during Permian (e.g., Pruner, 1987; Xu et al., 1997; Zhao et al., 2013).

5.2. Tectonic Implications

5.2.1. Closure of the Paleo-Asian Ocean

Timing for the MOB-NCC collision is essential for the evolution of the southeastern segment of Paleo-Asian Ocean. From comparison of Devonian poles of the MOB and the NCC, Kravchinsky, Sorokin, and Courtillot (2002) found negligible paleolatitude difference between the two blocks, indicating that the MOB was very close to the NCC in this time period. Besides, most previous paleomagnetic results support similar paleolatitudes for the MOB and the NCC during the Early Permian (Figure 8; Pruner, 1992; Chen et al., 1997; Li et al., 2012; Zhao et al., 2013), which leads these authors to propose a welded MOB-NCC at least since the Early Permian. However, recent paleomagnetic results suggested that previous data from the MOB that were mainly obtained from clastic rocks might be affected by inclination shallowing (Zhang et al., 2018). By adjustment of the inclination shallowing with an optimal flattening factor of $f = 0.6$, Zhang et al. (2018) placed the MOB at a relatively high paleolatitude of 30.8–32.0°N from Late Carboniferous to Late Permian and proposed that the MOB was still separated from the NCC by the Paleo-Asian Ocean during the Early Permian. However, with the same flattening factor, the Permian paleolatitude of the NCC increased from 21.5°N to 30.2°N (Zhang et al., 2018), similar as that of the MOB. This means that inclination shallowing of paleomagnetic data does not affect the relative position of the NCC and MOB, and the NCC

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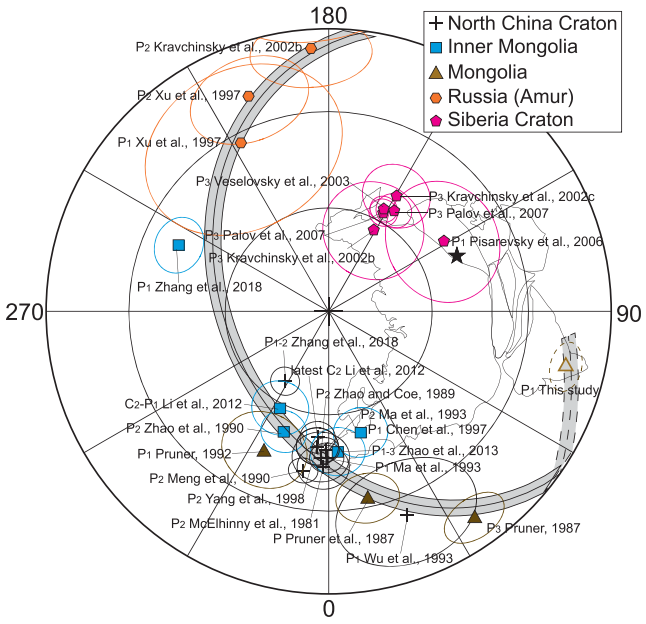


Figure 7. Equal-area projection of published Permian poles from the NCC, MOB, and SIB. Poles from the MOB are aligned along a small circle centered at the sampling location and this small circle transects Permian poles from the NCC. Permian poles from the SIB were far away from those of the NCC and the MOB, suggesting a wide Mongol-Okhotsk Ocean between the SIB and the NCC+MOB during the Permian.

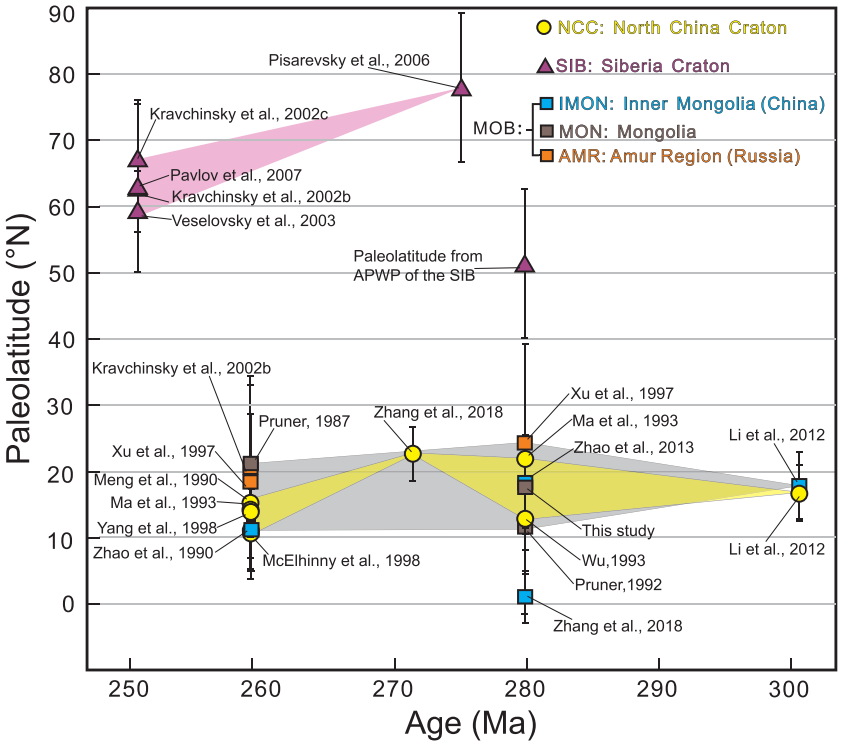


Figure 8. Paleolatitude comparison of latest Carboniferous and Permian paleomagnetic data from the NCC, MOB and SIB. Paleolatitudes of these three blocks are calculated from the data in Table 2. The 280 Ma paleolatitude of the SIB is calculated from compiled Early Permian poles in the APWP of SIB (Cocks & Torsvik, 2007; Metelkin et al., 2012; Smethurst et al., 1998).

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Figure 9. Paleogeographic reconstruction of the major blocks of northern Asia in the Early Permian on the basis of palaeomagnetic data from Table 2. Black star: palaeomagnetic sampling locality of this study. In the Early Permian, Tarim Craton has been welded with Eurasia Continent (Kazakhstan Block), whereas the connection between TAR and NCC-MOB is not clear. Paleomagnetic data suggest a large paleolatitude difference between NCC-MOB and SIB during Early Permian. This paleolatitude difference can be ascribed to the Mongol-Okhotsk Ocean. Bidirectional subduction of the Mongol-Okhotsk oceanic crust beneath both the southern margin (present coordinate) of SIB and northern margin of MOB can be illustrated by arc magmatic rocks along both continental margins. Abbreviations for blocks (using present-day boundaries) are EUR, European Craton; SIB, Siberia Craton; KAZ, Kazakhstan Block; MON, Mongolia Block; NCC, North China Craton; and TAR, Tarim Craton.

and MOB have the similar paleolatitude during the Permian. Moreover, several Permian poles with high paleolatitude (45°N–62°N) were obtained from southwestern Mongolia, and whether they can represent the MOB or not is still controversial. Our new Early Permian data were acquired from volcanic rocks that did not suffer from inclination shallowing, as clastic strata do. Therefore, the new Early Permian pole reported here can be considered as a better representative for the MOB. It reveals a consistent paleolatitude of the MOB and the NCC, indicating that the MOB should have been welded with the NCC during the Early Permian or was at least close to it. It also means that the Paleo-Asian Ocean between the MOB and the NCC should have been closed before the Early Permian.

Paleomagnetic results are supported by geological evidence. Sedimentological investigations in central Inner Mongolia showed that Permo-Carboniferous sedimentary rocks were deposited above older suture zones, representing a stable sedimentary cover after the previous orogenic event (Zhao et al., 2016). Based on sedimentological investigations in Inner Mongolia, Mueller et al. (1991) proposed that there was no oceanic basin in the Inner Mongolia region since the Late Carboniferous and if there was oceanic closure in late Carboniferous and younger times, the oceanic basin might have been farther north of Inner Mongolia. New sedimentological studies reveal that the Permo-Carboniferous strata in southern Mongolia and northern Inner Mongolia are mainly continental facies deposits with the same plant fossils (Zhao, Xu, & Zhang, 2017), accompanying widespread peralkaline-alkaline granitic rocks and bimodal volcanic rocks (Tong et al., 2015; Yarmolyuk et al., 2008). These rock and fossil assemblages indicate that a continental rift system should have existed in the southern MOB during Permo-Carboniferous (Zhao, Xu, & Zhang, 2017). In this case, the oceanic basin should be located to the north of southern Mongolia. Since paleomagnetic data show consistent paleolatitudes for northern Mongolia, Inner Mongolia, and the NCC (Figure 8), we may further conclude that no big oceanic basin existed from the NCC to northern Mongolia since the Early Permian. In other words, the oceanic basin should be that of the Mongol-Okhotsk Ocean to the north of the MOB that caused the large paleolatitude difference between the MOB and the SIB (Figure 8). Thus, the Paleo-Asian Ocean between the NCC and the MOB should have been closed at least since the Early Permian, forming a

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unified North China-Mongolia Block. Note that remnant seas were still in existence within the MOB during Permian as indicated by the Permian marine strata distributed in southern Mongolia and Inner Mongolia (Johnson, 2004; Zhao et al., 2016). These marine basins can be ascribed to postorogenic extension, which caused intensive Early Permian extension-related magmatism, represented by ubiquitous bimodal volcanic rocks and granitic intrusions (Tong et al., 2015; Zhao, Xu, & Zhang, 2017).

5.2.2. Early Stage of the Mongol-Okhotsk Ocean

Most research works of the Mongol-Okhotsk Ocean are focused on the closure of different segments of this 3,000-km-long orogenic belt, whereas the early stage evolution of this belt is poorly understood. The western part of the Mongol-Okhotsk Belt recorded its early stage evolution, where Devonian and Carboniferous clastic turbidite deposits are widely distributed (Zonenshain et al., 1990). Meanwhile, two ophiolitic mélanges with zircon U-Pb ages of 325–314 Ma have been reported from the Adaatsag and Khuhu Davaa regions in central Mongolia, representing remnants of the Mongol-Okhotsk oceanic crust (Tomurtogoo et al., 2005; Zhu et al., 2018). By reviewing geological and geochemical studies on the Late Paleozoic-Mesozoic magmatic rocks on the northern side of the Mongol-Okhotsk suture zone, Donskaya et al. (2013) suggested that the northward Mongol-Okhotsk oceanic subduction began in the Devonian and reached its climax in the Permian-Triassic. The subduction was revealed by the middle Carboniferous-Triassic Selenge Arc (Donskaya et al., 2013). On the southern side of this belt, Zhao, Xu, and Jahn (2017) reported three Permian (273–254 Ma) subduction-related granitic massifs in eastern Mongolia. By compilation of published age data of Permian and Triassic magmatic rocks from northeastern Mongolia and northeastern China, Zhao, Xu, and Jahn (2017) proposed a Permian-Triassic magmatic pulse along the north margin of the MOB. Therefore, subduction of the Mongol-Okhotsk oceanic crust beneath both the southern SIB (present coordinate) and the northern MOB is indicated during Permian-Triassic (Figure 9).

For the Permo-Carboniferous period, paleomagnetic studies on the SIB mainly focused on the latest Permian (ca. 251 Ma) Siberian basaltic trap (Table 2). Consistent Late Permian poles have been obtained from different parts of the SIB (Table 2 and Figure 7). Unlike the large data set of Late Permian poles, only one Early Permian (275 $\pm$ 4 Ma) paleomagnetic pole was obtained from mafic dykes at the southern margin of the SIB near the Baikal Lake (Pisarevsky et al., 2006). Sole reverse polarity that is consistent with the PCRS, and a positive baked contact test was used to argue for a primary remanence (Pisarevsky et al., 2006). This Early Permian pole indicates a high paleolatitude at ~79°N for the SIB during the Early Permian (Figure 8). However, this high paleolatitude is conflicting with the Early Permian paleogeographic reconstruction of the SIB based on its APWP, in which the SIB is located at a paleolatitude of 40°–65° (Figure 8; e.g., Smethurst et al., 1998; Cocks & Torsvik, 2007; Metelkin et al., 2012). This difference leads Pisarevsky et al. (2006) to propose that the SIB was not connected with Laurasia by 275 Ma. The discussion of this difference is beyond the scope of this paper. Here we use the Early Permian pole from the APWP of the SIB for paleogeographic reconstruction (Figure 9). Considering the low paleolatitude of the MOB (~12°–24°), there was a large (~30°) latitudinal difference between the MOB and the SIB during the Early Permian, and this latitudinal difference increased in Late Permian due to the northward migration of the SIB (Figure 8; Smethurst et al., 1998; Cocks & Torsvik, 2007). This latitudinal difference can be interpreted by the Mongol-Okhotsk Ocean between the SIB and the MOB (Figure 9).

Note that the Kazakhstan Block had joined Laurussia by the Uralian Orogeny during the Late Carboniferous to earliest Permian (Cocks & Torsvik, 2007), and the SIB welded with the Kazakhstan Block in the Late Carboniferous-Early Permian, forming the proto-Eurasia continent (Abrajevitch et al., 2008; Filippova et al., 2001). Paleomagnetic data show that the southern margin of the proto-Eurasia continent (present shape and position) was longitude-parallel during the Permian (Figure 9; Kravchinsky, Cogné, et al., 2002; Van der Voo et al., 2015). In contrast, the northern margin of the MOB faced to the northeast (Figure 9; Zhao et al., 2013). In this case, there is a >100° intersection angle between the western and southern margins of the Mongol-Okhotsk Ocean (Figure 9). This kind of paleogeographic pattern resulted in the proposal that subduction along the two continental margins might belong to one subduction belt (e.g., Zorin, 1999). Meanwhile, geological studies suggest that the Tarim Craton has collided with the Kazakhstan Block during the Late Carboniferous (e.g., B. Han et al., 2011), but the connection between the Tarim Craton and the NCC-MOB during the Early Permian is still unclear (Figure 9). Although this problem is out of scope of this study, it relates to an important issue of whether the Mongol-Okhotsk Ocean was connected with the Paleo-Tethys

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Ocean during Permian (Enkin et al., 1992; Gilder et al., 2008; Van der Voo et al., 2015; Wu et al., 2017). More geological and paleomagnetic studies are needed to uncover this problem.

5.2.3. Intracontinental Strike Slip

All paleomagnetic poles obtained from the MOB distributed along a small circle (Figure 7), indicating that the unified NCC-MOB is not a rigid block, which can be ascribed to posterior intracontinental vertical-axis strike-slip-related rotations, especially along previous suture zones (e.g., Zhao et al., 2015). The strike-slip-related rotations have been reported by previous paleomagnetic and structural studies (Enkin et al., 1992; Metelkin et al., 2010; Zhao et al., 2015), represented by Mesozoic shear zones within and along the margin of the MOB (e.g., Xilamulun Fault Zone and East Gobi Fault Zone; Webb et al., 2010; Zhao et al., 2015). It is noted that paleomagnetic poles of the MOB from Inner Mongolia part overlap with poles from the NCC, indicating that no significant rotation occurred for this part since Permian. However, poles of the MOB from the Mongolia part show different degrees of anticlockwise rotation relative to poles of the NCC (Figure 7), which may be induced by the sinistral shear along the East Gobi Fault Zone (Webb et al., 2010). On the contrary, poles of the MOB from the Amur region in Russia show a large degree of clockwise rotations relative to poles of the NCC (Figure 7), which might be ascribed to the fact that sample locations for these three poles are all located inside of the Mongol-Okhotsk suture zone. Post-Jurassic strike-slip movement along this suture zone may cause significant internal rotations. Detailed studies on the East Gobi Fault Zone and Xilamulun Fault Zone have been performed, which shows Late Triassic strike-slip movements of these faults that were triggered by far-field stress effect due to the Triassic collision between the NCC and the South China Craton (e.g., Zhao et al., 2015), and coeval closure of the western part of the Mongol-Okhotsk Ocean (e.g., Metelkin et al., 2010; Zhao et al., 2015).

6. Conclusions

In order to uncover the Late Paleozoic paleogeographic position of the MOB and further understand evolution of both the Paleo-Asian Ocean and the Mongol-Okhotsk Ocean, a paleomagnetic study is performed on the Early Permian volcanic strata in the Bayandun region of northeastern Mongolia. Zircon U-Pb dating results reveal an Early Permian age (283 ± 3 Ma) for the studied volcanic strata. Magnetic mineralogy investigations reveal magnetite and Ti-magnetite as the main magnetic carriers. Magnetic directional isolation displays both primary and secondary magnetizations. ChRM directions isolated from seven sites show consistent reverse polarity, corresponding to the Permo-Carboniferous (Kiaman) Reverse Superchron. Site-mean directions pass fold tests, and an Early Permian paleomagnetic pole is calculated for the MOB at $\lambda/\varphi = -14.9°\text{N}/76.8°\text{E}$ ($A_{95} = 5.7°$) with $N = 7$ sites. All published Permian paleomagnetic poles from the MOB are aligned along a small circle with the sampling location as center, and this small circle intersects Permian poles from the NCC but far away from those from the SIB. These characteristics indicate that (1) the MOB should have welded with the NCC before the Early Permian or was at least close to it. (2) The welded MOB-NCC was separated from the SIB by the Mongol-Okhotsk Ocean with ~30° latitudinal difference in the Early Permian. (3) Significant vertical-axis strike-slip-related rotations occurred within and along the margins of the unified MOB-NCC due to far-field stress effect produced by posterior orogenic events.

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